FLYRANK INTERNSHIP · FRONTEND AI ENGINEERING TRACK · BUILD PHASE (CORE)

# FE-06: Streaming AI Chat Interface

Production Architecture: Server-Sent Events (SSE) Route Handler, Centralized Model Configuration, Flicker-Free Thinking Shimmer Handoff, State-Preserving Stop Button, and User-Aware Auto-Scroll Pinning.

| CANDIDATE | ASSIGNMENT CODE | AI DEVELOPMENT ENGINE | ASSIGNED WORKLOAD |
|---|---|---|---|
| **KM Siam** | **FE-06 (Week 4)** | **Google Antigravity** | **6 Hours · Core Build** |

## 1. Executive Summary: The Defining UI Pattern of 2026

Token-by-token streaming conversation is the defining user interface paradigm of the modern AI era. **FE-06** delivers a hardened, production-grade streaming AI chat interface built on Next.js 15 App Router, React 19, and Tailwind CSS.

The system decouples server-side inference governance from client-side reactive rendering:

- **Server-Side Route Handler ( `src/app/api/chat/route.js` ):** Communicates via Server-Sent Events (SSE) emitting `text-delta` frames, while keeping API credentials strictly server-side.
- **Centralized Model Config ( `src/lib/aiConfig.js` ):** Encapsulates system prompts, model parameters (Claude 3.7 Sonnet, Gemini 3.8 Flash High, GPT-4o), and deterministic streaming fallbacks.
- **Client Streaming UI ( `src/components/chat/StreamingChat.jsx` ):** Features thinking indicator handoff, abort control with partial message retention, intelligent auto-scroll, and multi-turn `localStorage` persistence.

## 2. Architecture & Mentor Invariants Verification

### 1. The Thinking Indicator Handoff

Before the first token arrives, a pulsing shimmer indicates active synthesis. When the first SSE chunk arrives, the UI executes a seamless state handoff (transitioning status from `'thinking'` to `'streaming'` ) without unmounting the message container, eliminating visual flicker.

### 2. State-Based Stop Button

Aborting is treated as a state machine transition, not a cosmetic UI toggle. Clicking "Stop" fires `AbortController.abort()` , halts stream ingestion, preserves whatever partial tokens were received, re-enables the input prompt field, and permits immediate subsequent user turns.

### 3. Resilient User-Aware Auto-Scroll

The chat container maintains an active pin ( `scrollHeight - scrollTop - clientHeight < 80px` ). The moment a user scrolls up to review prior answers, the auto-scroll lock is released, and a floating "Scroll to latest" button appears, preventing disruptive jumping.

### 4. Multi-Turn History Persistence

Conversations persist to `localStorage` ( `flyrank_streaming_chat_history_v1` ). A browser reload or accidental tab closure restores the full conversational thread with intact token metrics.

## 3. Server Route Handler & Protocol Specification (SSE)

```
POST /api/chat HTTP/1.1
Host: flyrank-internship-front-end-ai-engineering.vercel.app
Content-Type: application/json

{
  "messages": [
    { "role": "user", "content": "Explain streaming chat architecture" }
  ],
  "model": "claude-3-7-sonnet"
}

HTTP/1.1 200 OK
Content-Type: text/event-stream; charset=utf-8
Cache-Control: no-cache, no-transform
Connection: keep-alive

data: {"type":"text-delta","text":"### Analysis"}

data: {"type":"text-delta","text":" & Architectural"}

data: {"type":"text-delta","text":" Breakdown"}

data: {"type":"finish","finishReason":"stop"}

data: [DONE]
```

## 4. Centralized Model Configuration Module ( `src/lib/aiConfig.js` )

```
export const MODEL_CONFIG = {
  provider: 'anthropic',
  defaultModel: 'claude-3-7-sonnet',
  supportedModels: [
    { id: 'claude-3-7-sonnet', name: 'Claude 3.7 Sonnet (Anthropic)', tokensPerSec: 42 },
    { id: 'gemini-3.8-flash', name: 'Gemini 3.8 Flash High (Google)', tokensPerSec: 68 },
    { id: 'gpt-4o', name: 'GPT-4o (OpenAI)', tokensPerSec: 38 }
  ],
  temperature: 0.7,
  maxOutputTokens: 2048,
  topP: 0.95
};
```

## 5. Comprehensive Automated Verification Suite (41/41 Passed)

```
$ npm test

> flyrank-internship-front-end-ai-engineering@1.0.0 test
> node tests/promptcraft.test.js && node tests/routes.test.js && node tests/accessibility.test.js && node tests/streaming.t

🚀 PromptCraft Suite: 10/10 passed
🚀 Routes & Skeleton Suite: 14/14 passed
🚀 Accessible Components Suite: 10/10 passed

🚀 Starting FE-06 Streaming AI Chat Interface Automated Test Suite...

--- 1. Centralized System Prompt & Model Configuration ---
  ✅ PASS: aiConfig.js exports SYSTEM_PROMPT with high-signal guidelines
  ✅ PASS: aiConfig.js exports MODEL_CONFIG with Claude, Gemini, and GPT options
  ✅ PASS: getSimulatedResponse returns non-empty structured response

--- 2. Server-Side Streaming Route Handler (POST /api/chat) ---
  ✅ PASS: POST /api/chat returns HTTP 200 with text/event-stream content type
```

```
   ✅ PASS: Streaming route delivers text-delta SSE chunks and finishes cleanly
   ✅ PASS: Streaming route respects AbortController signal without throwing

--- 3. Client Component Architecture & Mentor Invariants ---
   ✅ PASS: StreamingChat.jsx implements thinking indicator, stop button, auto-scroll, and persistence

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Summary: 7/7 FE-06 tests passed · Total Suite: 41/41 passed (100%)
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$ npm run build
> next build
   ▲ Next.js 15.5.29
 ✓ Compiled successfully in 4.5s
 ✓ Generating static pages (14/14)
   ├ ƒ /api/chat
   ├ ○ /chat
```

FlyRank Frontend AI Engineering · FE-06 Assignment Deliverable

Candidate: KM Siam (k.m.siam2019@gmail.com) · Southeast University

Engine: Google Antigravity (Gemini 3.8 Flash High)